



Short communication

Effects of high winds on a long-span sea-crossing bridge based on structural health monitoring

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ABSTRACT

This paper focuses on the effects of high winds on vibrational responses and modal parameter's variation for a unique sea-crossing cable-stayed bridge, i.e. Donghai Bridge, based on the long-term structural health monitoring (SHM) data. This bridge is located in a typhoon-prone area of the northwestern Pacific Ocean, and 80% of the vehicles traversing it are heavy-load container trucks. The relative influence of high winds and heavy-load traffic on the structural vibration responses is investigated by a comparison of typical operational cases. The high wind dominates in the girder's lateral vibration of Donghai Bridge, while the girder's vertical and torsional accelerations are more sensitive to heavy-load traffic rather than high winds. Also, the frequency band of the wind excitation is lower than that of traffic load. During high winds, the modal parameters of this bridge are relatively difficult to stably identify, thereby leading to a well-pronounced discreteness of the parameters' estimates. This paper could serve as a field evidence for the wind-resistant design and the performance evaluation of bridges in similar operational conditions.

1. Introduction

Wind is one of the dominant loads on large-span bridges. Accordingly, bridge wind engineering has increasingly gained attention with the increase of the number of bridges across seas and wide rivers. The development and practice of the structural health monitoring (SHM) technology provide an in-situ data platform for the wind-resistance study on long-span bridges. Compared with numerical simulation and wind tunnel experiments, field measurement is a more straightforward and effective way to determine the on-site wind characteristics and provide the wind-induced responses of full-scale bridges. In recent years, an increasing amount of research is conducted in relation to the wind characteristics and wind effects around and on long-span bridges based on the SHM data. Case studies include the Tsing Ma Bridge (Xu, 2013), Akashi Kaikyo Bridge (Miyata et al., 2002), Runyang Suspension Bridge (Wang et al., 2016a), Humber Bridge (Rahbari et al., 2015), Ting Kau Bridge (Ni et al., 2015), Sutong Bridge (Wang et al., 2016b), Dongting Lake Bridge (Ni et al., 2007), Xihoumen Bridge (Li et al., 2014), Jiubao Bridge (Wang et al., 2014), Chongqi Bridge (Sun et al., 2017), Taizhou Bridge (Tao et al., 2016), and so on.

Each bridge has its own unique structural configuration and local topography, which may result in huge variability among the field

measurement of different bridges. Thus, separate studies are still required for individual cases. Employing six years of monitoring data from Donghai Bridge at Shanghai, the authors in this study investigate the effects of high winds on structural vibration as well as the variation in the modal parameters. As China's first offshore sea-crossing bridge, Donghai Bridge is located in the typhoon-prone area of the northwestern Pacific Ocean. Compared to ordinary inland bridges, the severe wind field in Donghai Bridge enhances its value for wind-resistance research. Moreover, this bridge remains as the only approach link to the Yangshan Deepwater Container Port, and 80% of the vehicles traversing the bridge are heavy-load container trucks, resulting in a higher level of traffic load than ordinary highway bridges. On account of significant effects of both high winds and heavy-load traffic, this study tries to compare the relative influence of high winds and heavy-load traffic on the vibrational responses. Under the field measurement context, the effects of these two kinds of dynamic loads are usually difficult to separate from each other. Fortunately, the authors found some typical operational cases of Donghai Bridge (i.e., absence of high winds with no traffic load, absence of high winds but with traffic load, presence of high winds but with no traffic load, and presence of high winds and with traffic load), thereby leading to an intuitive separation of the wind and traffic effects.

The next section of this paper contains a brief background

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introduction to this super infrastructure and its monitoring system. Sections 3 and 4 deliberate on the high wind effects through a comparison with the traffic effects in terms of the magnitude and the frequency range of the girder's acceleration. A discussion on the influence of high winds on the variation of the modal parameters of Donghai Bridge is presented in Section 5. Finally, Section 6 summarizes the paper.

2. Engineering background and high wind process

Opened on December 10, 2005, Donghai Bridge is the vital linkage between Shanghai and the Yangshan Deepwater Container Port with a full length of 32.5 km. The main navigational opening is a five-span and twin-pylon cable-stayed bridge with a 420-m center span (Fig. 1). The girder is a three-cell, steel-concrete composite box girder consisting of steel bottom flanges and webs and pre-stressed, high-performance concrete top flanges with a width of 33 m and a depth of 4 m (Fig. 2). The pylons are 150 m tall and feature an inverted Y shape. The stays, which are pre-fabricated parallel wire strands (PPWSs), are twinned and arranged in a single cable plane. Twenty-four pairs of stays per girder cantilever are provided, with the longest one being 227 m in length.

To facilitate the management and maintenance of the bridge, an advanced structural health monitoring system (SHMS) was implemented in September 2006 to monitor the environmental/operational actions and the static/dynamic responses of Donghai Bridge in real time. Fig. 1 shows a schematic distribution of the measuring points for the wind speed/direction, temperature, and acceleration. The two pylons are equipped with five monitoring sections along the height. Three accelerometers are mounted at the bottom of each pylon (elevation of 9 m) to measure the vibration in three orthogonal direction; the sections at the elevation of 51.5 m, 102.5 m, and 152.5 m are respectively installed with two accelerometers to measure the longitudinal and transversal vibration of the pylon; and the north pylon also has one anemometer at the pylon's top (elevation of 159 m) and eight thermometers at the elevation of 102.5 m. The girder is equipped with ten monitoring sections in the longitudinal direction. The sections 1, 3, 5, 7, 9, and 10 contain three accelerometers, two vertical and one lateral measuring points; the sections 2, 4, and 8 are set with six or 18 thermometers; and section 6 at the mid-span has four accelerometers (two vertical, one lateral, and one longitudinal measuring points), 20 thermometers and one anemometer.

Moreover, the eight longest cables connected to the south pylon are selected for monitoring, each with an accelerometer and a thermometer. The sampling frequency of the thermometers, accelerometers, and anemometers is 1 Hz, 50 Hz, and 1 Hz, respectively.

This study focuses on the effects of high winds. Over the period 2007–2012, Donghai Bridge experienced several typhoon events. The bridge was closed to traffic during typhoons Krosa, Muifa, Haikui, and Bolaven, which provided an opportunity for us to fully separate the effect of wind and that of the traffic load. Aside from the typhoon periods, Donghai Bridge was also closed on February 22, 2011 when a major traffic accident occurred on the bridge. Also, the maximum instantaneous wind speed during these six years was recorded during a severely convective weather of short duration on August 22, 2008 (hereinafter referred to as the August 22 Storm) (Zhou et al., 2014). These two periods will also be selected for further analyses. Fig. 3 shows the time histories of the instantaneous wind speed/direction at the deck level during the periods mentioned above. In Fig. 3(f), it is noted that no continuous wind direction can be observed because of the low wind speed.

3. Comparison between high wind and traffic effects: response amplitude

The measurements in the bridge closure period during typhoons were firstly selected to investigate the wind effects separately. The data measured at the vertical acceleration measuring points (5AV007 and 5AV008 in Fig. 2) and the lateral acceleration measuring point (5AB001 in Fig. 2) are converted into vertical acceleration $\ddot{V}$, lateral acceleration $\ddot{H}$, and torsional (angular) acceleration $\ddot{\alpha}$ at the centroid location of the mid-span cross-section. Fig. 4 shows the correlation of the root mean square (RMS) of the centroid vertical acceleration $\ddot{V}$ and the absolute values of the 10 min mean speed of the orthogonal crosswind (the projected wind speeds perpendicular to the longitudinal axis of the bridge) according to the bridge closure data for the three typhoon events (i.e., Krosa, Muifa, and Bolaven). The girder's acceleration does not approach zero in a low wind speed. However, a roughly quadratic curve relation exists between them when the orthogonal crosswind speed exceeds 13 m/s, which may be attributed to the fact that the wind pressure is proportional to the square of the wind speed.

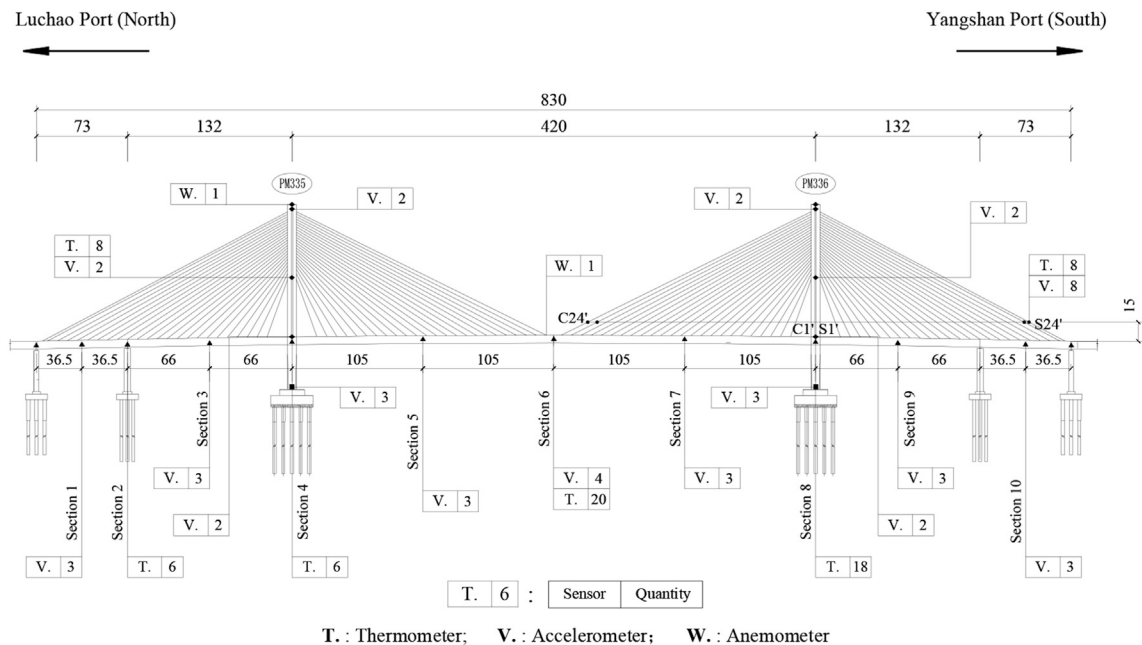


Fig. 1. Elevation view of Donghai Bridge and schematic distribution of the measuring points for the wind speed/direction, temperature, and acceleration (unit: m).

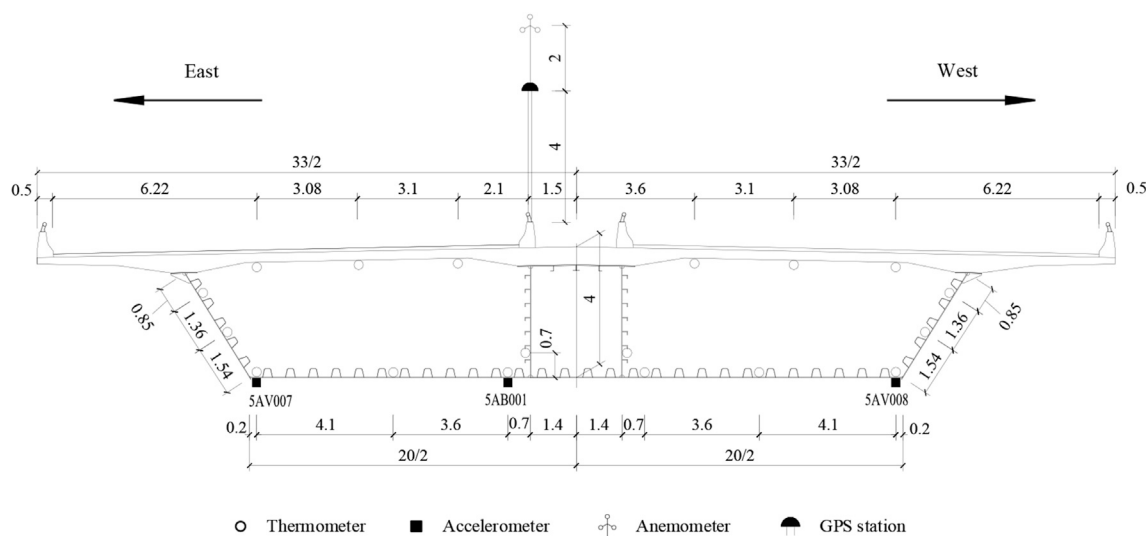


Fig. 2. Schematic distribution of the measuring points in the mid-span cross-section of the girder (unit: m).

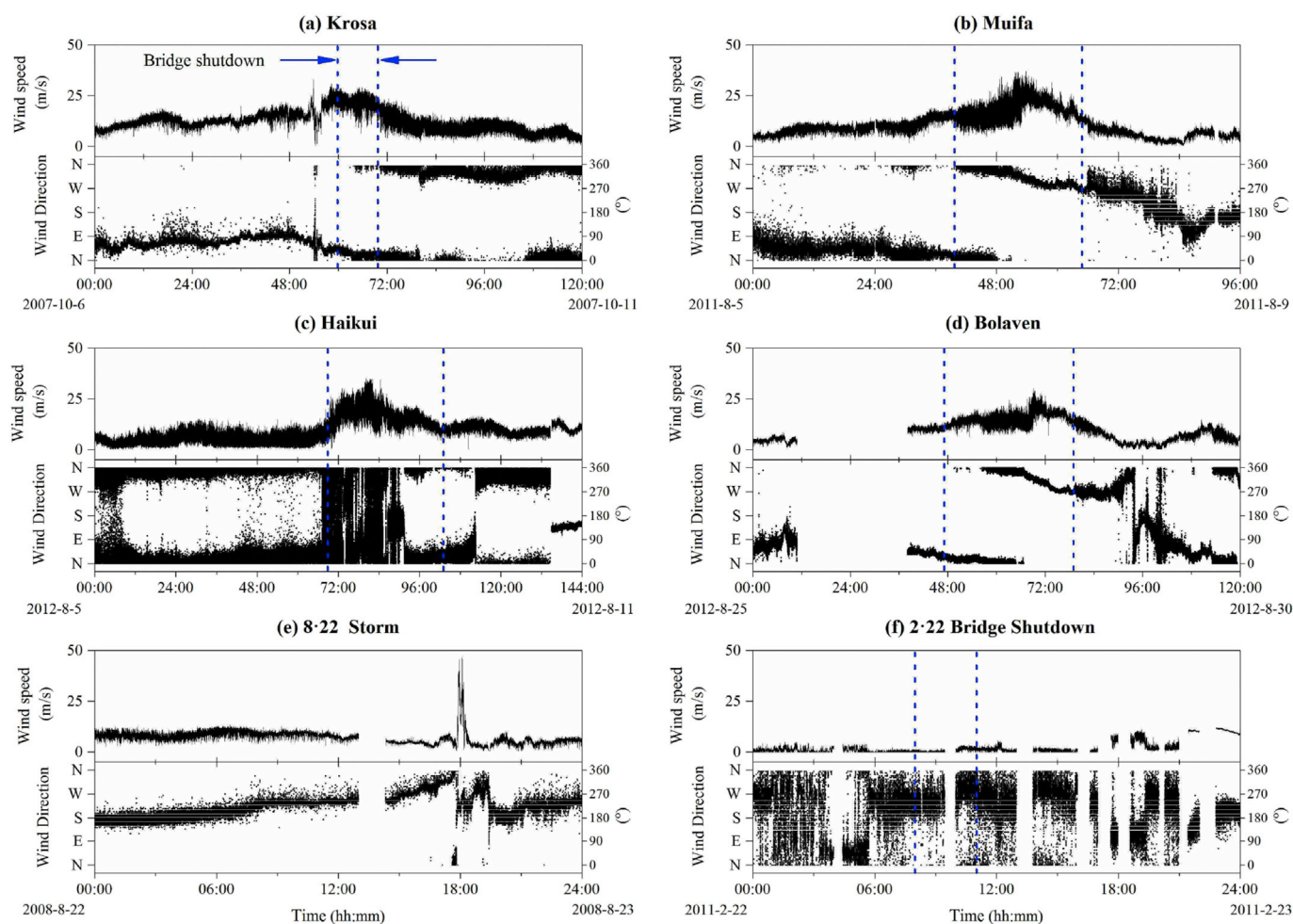


Fig. 3. Time history of wind speed/direction at the bridge site during the selected periods.

Since wind and traffic loads constitute two major sources of the dynamic responses of bridges, we compared the effects of high winds and heavy-load traffic on the girder's acceleration amplitude of Donghai Bridge. Fig. 5 shows the time history of the mid-span acceleration response during Typhoon Haikui, the most violent typhoon in the period

2007–2012. The structural acceleration response in the bridge closure period is dramatically different from that in the normal traffic period. Clearly, both high winds and heavy-load traffic have a non-negligible influence on the girder vibration.

Fig. 6 further shows comparisons of the instantaneous maximum

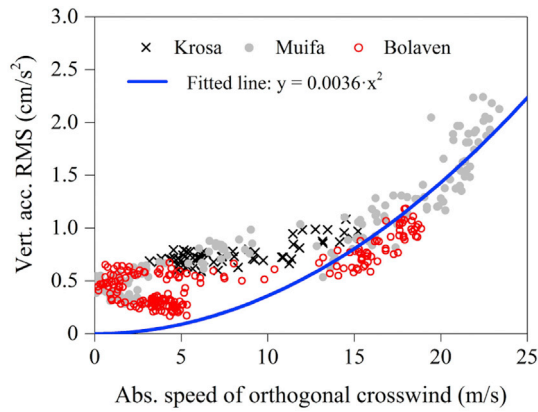


Fig. 4. Vertical vibration versus wind speed.

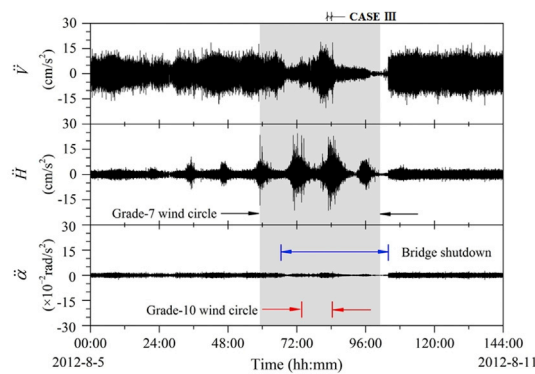
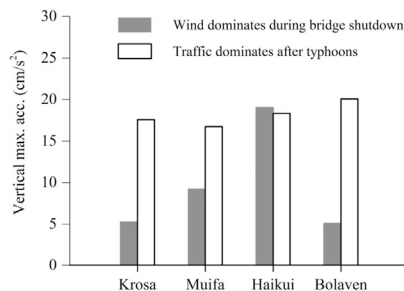


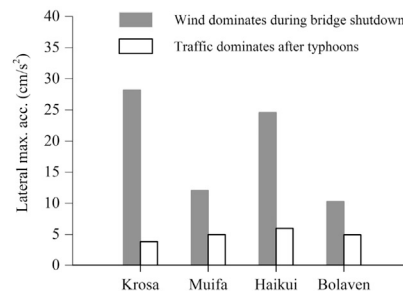
Fig. 5. Acceleration responses at mid-span of the girder during the Haikui typhoon event.

acceleration at mid-span of Donghai Bridge in two different periods: (1) during the bridge closure periods of four typhoon events, when there is the presence of winds with no traffic load and (2) during normal traffic periods, when there is the absence of winds but with traffic load. The traffic load is the dominant factor for the girder's vertical and torsional vibration because only such an extremely strong typhoon as Haikui can excite the girder's vertical/torsional acceleration to the same level as the daily traffic load. However, the dominant factor for the lateral vibration of Donghai Bridge is the wind load. The lateral acceleration amplitude in typhoon events is equal to several times that of the daily traffic load, which is understandable for the lateral vibration not being coincidental with the principal acting direction of traffics.

Noted that the sensitivity of the structural vertical/torsional vibration to the wind and traffic load depends on the structural configuration and the relative intensity of actions. Unlike that of Donghai Bridge, the



(a) Vertical



(b) Lateral

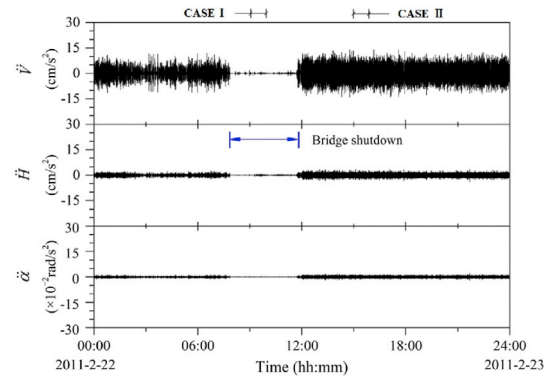


Fig. 7. Mid-span acceleration response of the girder of Donghai Bridge, February 22, 2011.

girder's vertical/torsional vibration of Shanghai Yangtze River Bridge, which is a 730 m central span cable-stayed bridge, is still dominated by the wind load (Zhou et al., 2017a). The traffic load of Shanghai Yangtze River Bridge is a conventional highway load, which is much lower than the container trucks featured traffics on Donghai Bridge.

4. Comparison between high wind and traffic effects: frequency content

Except for the response amplitude, the frequency contents of the structural vibration induced by wind and traffic load are compared. This section explores the following four 1 h data sets corresponding to the four cases of wind and traffic load combination:

1. CASE I, absence of high winds with no traffic load: 2011-2-22 9:00–10:00 (bridge closure because of a traffic accident).
2. CASE II, absence of high winds but with stochastic traffic load: 2011-2-22 15:00–16:00 (traffic resumption after the accident).
3. CASE III, presence of high winds but with no traffic load: 2012-8-8 11:00–12:00 (bridge closure during Typhoon Haikui).
4. CASE IV, presence of high winds and with stochastic traffic load: 2008-8-22 17:30–18:30 (August 22 Storm period).

As the temperature remained stable, and no high wind was observed at the bridge site during the bridge closure period (Fig. 3(f)), CASE I can serve as the baseline case that has neither wind nor traffic loads. The frequency band characteristics of excitations exerted by the wind and the traffics can be studied by comparing the baseline case with the other three operational cases. Figs. 7 (CASEs I and II), 5 (CASE III), and 8 (CASE IV) show the time histories of the girder's acceleration at the mid-span for the four cases.

The power spectral densities (PSDs) of the vertical, lateral, and torsional acceleration at the girder's mid-span under CASEs II, III, and IV

Fig. 6. Comparison of the instantaneous maximum acceleration at mid-span of the girder between the bridge closure and post-typhoon periods.

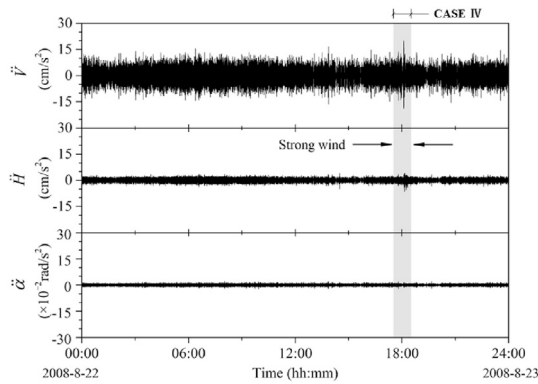


Fig. 8. Mid-span acceleration response of the girder of Donghai Bridge, August 22, 2008.

are compared with those under CASE I (Figs. 9–11). The cut-off frequency in these figures is 25 Hz, for the sampling frequency of the acceleration signals being 50 Hz. Focusing on the PSD curves within the 0–5 Hz range, the following findings can be observed: (1) referring to the comparison of CASES I and II, the acceleration PSD significantly increases within the frequency band of approximately 2.5–5.0 Hz when only the traffic load exists; (2) referring to the comparison of CASES I and III, the PSD that is at the range of approximately 0–2.5 Hz obviously increases when only the high winds exist; and (3) referring to the comparison of CASES I and IV, the PSD within the range 0–5 Hz has a relatively even increase when both the wind and the traffics exist. These results indicate that the exciting frequency of the traffic load is higher than that of the strong wind. Table 1 presents the mean square values (square of the RMS value, which represents the signal's energy and is equal to the area below the PSD curve) within different frequency bands under the four typical operational cases. The vibration energy within 2.5–5.0 Hz exceeds that in the 0–2.5 Hz range in the absence of high winds with traffic load. Meanwhile, the relative relation of the vibration energy in these two frequency bands reverses in the presence of high winds with no traffic load. The vibration energy within these two ranges also remains approximately at the same level in the presence of high winds with stochastic traffic load. The qualitative relationship presented in Table 1 is consistent with the observations in the PSD plots. In addition, the peaks of the PSD curves under CASES III and IV in Figs. 9–11 deviate leftward with respect to those under CASES I and II, especially for the high-order modes, which means that the modal frequencies of Donghai Bridge under CASES III and IV tend to decrease. Considering that CASES I and II occurred in February and CASES III and IV occurred in August, the decreasing modal frequencies

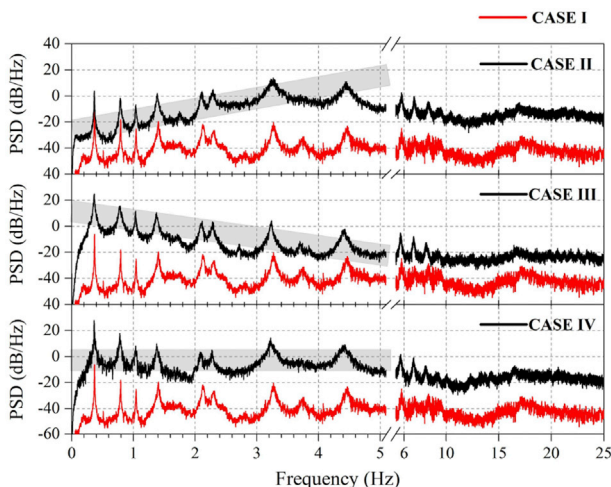


Fig. 9. PSD comparison of the vertical acceleration at mid-span.

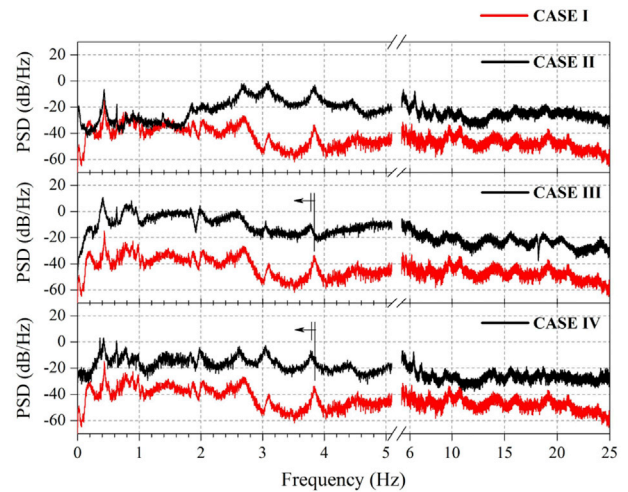


Fig. 10. PSD comparison of the lateral acceleration at mid-span.

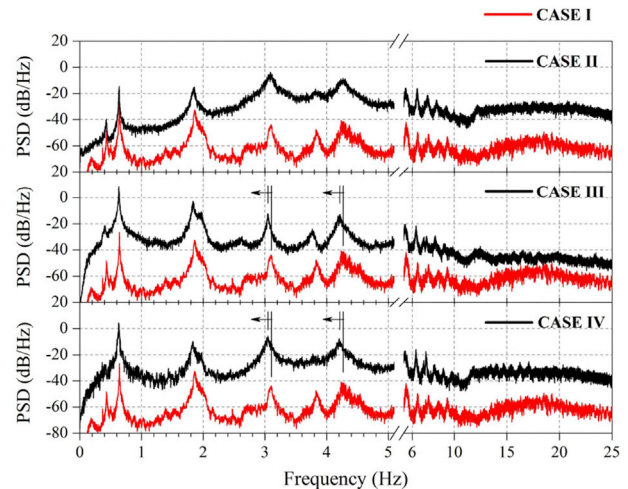


Fig. 11. PSD comparison of the torsional angular acceleration at mid-span.

can be explained in terms of the rising temperature.

In addition to the girder, the accelerations at the bottom of the pylon (or the top of the foundation platform) are also analyzed. The PSD plot in Fig. 12 shows that the frequency bands of the accelerations at the pylon's bottom induced by high winds are also lower than those induced by the traffic load. Moreover, the results of CASE II in Fig. 12 demonstrate that the traffic-induced vibration can also be transferred to the pylon's bottom even if the traffic load directly acts upon the girder. For example, the RMS of the lateral acceleration at the north pylon's bottom under CASES I and II are 0.0043 cm/s² and 0.0573 cm/s², respectively. The latter is approximately 13 times that of the former. Undoubtedly, the traffic load excitation is much weaker than that of frequent earthquakes. Under such a low dynamic loading level, however, the vibration responses at the bottom of pylons are significantly different from those under traffic-free conditions. Hence, for the dynamic modeling of Donghai Bridge, especially in the case of time history analyses, it might not be appropriate to specify the pylon's bottom as a fixed boundary. Instead, it is advisable to introduce the elastic constraints at the bottom of pylons.

As mentioned in Section 3, the amplitudes of the girder's vertical and torsional acceleration are dominated by the traffic load when the wind and traffics simultaneously act on Donghai Bridge. Therefore, the wind effects would be masked, and the correlation between the wind speed and the acceleration RMS would become quite weak (Fig. 13(a), where the data span from 2007 to 2012 and each point corresponds to a time

Table 1

Statistics of the mean square values for the mid-span acceleration responses.

Band (Hz)	Vertical (cm/s ²) ²		Lateral (cm/s ²) ²		Torsional ($\times 10^{-2}$ rad/s ²) ²	
	[0, 2.5]	[2.5, 5.0]	[0, 2.5]	[2.5, 5.0]	[0, 2.5]	[2.5, 5.0]
CASE I: absence of high winds with no traffic load	0.002	0.001	0.002	2.39E-04	2.82E-05	9.13E-06
CASE II: absence of high winds with traffic load	0.347	3.686	0.013	0.179	0.002	0.055
CASE III: presence of high winds with no traffic load	6.487	0.156	1.647	0.236	0.081	0.006
CASE IV: presence of high winds with traffic load	3.908	2.377	0.140	0.091	0.031	0.031

interval of 1 h). However, the analysis in this section shows that the vibration acceleration in the low frequency band should correlate well with the wind (Fig. 13(b)). In Fig. 13(b), the scatter plot of the mid-span vertical acceleration RMS in the frequency range 0–2.5 Hz versus the absolute speed of the orthogonal crosswind reveals a similar quadratic curve relationship as in Fig. 4. In addition, it can also be observed on Shanghai Yangtze River Bridge that the frequency band of the wind-induced vibration is lower than that related to the traffic load. It is probably a universal phenomenon that the girder's vibration increases for the frequency contents of approximately 2 Hz as the vehicles traverse the bridge. In the case of the slight differences in such factors as the pavement surface roughness and vehicle speed, the frequency range of excitation exerted by the traffic load is mainly correlated to the natural frequency of vehicles. For the purpose of driving comfort, the first-order vertical vibration frequency differs very slightly among different types of vehicles (e.g., the vertical fundamental frequency for ordinary trucks normally ranges from 1.5 Hz to 2.0 Hz, that of passenger buses ranges from 1.2 Hz to 1.8 Hz, and that of limousine cars ranges from 1.0 Hz to 1.3 Hz (Zhang, 2012)). As a result, the traffic-induced vibration should have roughly similar frequency contents.

5. Effects of high winds on the variation of the modal parameters

Modal parameters are one of the key indicators representing the structural overall performance. Their variation is the focus of bridge monitoring. This section investigates the strong wind effects on the variation of the modal frequency and the damping ratio of the first-order vertical bending, lateral bending, and torsional modes of the girder. These modal parameters were identified through the data-driven stochastic subspace identification (SSI-Data) method with a total of 40 acceleration measuring points distributed across the bridge. The results of the modal identification would vary with the choice of algorithm parameters; hence, we tried to reduce the estimation uncertainty by averaging multiple identification results for a specific data set. The following

steps were adopted (Ubertini et al., 2013): (1) the raw data were downsampled to the frequency of 5 Hz after the anti-aliasing filtering; (2) half the number, i , of the block rows of the Hankel matrix was increased from 40 to 60 with a step of five, and the system order, n , was increased from 90 to 150 with increments of 10. The SSI-Data algorithm was executed once for each pair (i , n), thereby resulting in a set of 35 estimates for each modal parameter; and (3) the final value of a parameter for a given data set took the average of the 35 estimates, which could be referred to as the within-sample average. Similarly, the standard deviation from such averages was called the within-sample standard deviation, which reflects the instantaneous variability of the modal parameter identification results at a given time or for a given data set. Fig. 14 illustrates the evolutions of each modal parameter (within-sample averages), mean wind speed at deck level, acceleration RMS at mid-span, and structural averaged temperatures during Typhoon Haikui with a 10 min sample interval. Fig. 15 shows the stability of the identification results over time, where the within-sample standard deviation of each modal parameter is further divided by the mean value of all within-sample averages during Typhoon Haikui, thereby obtaining the normalized standard deviation δ .

Although no quantitative variation trend of the modal parameters is observed during high winds (Fig. 14), the dimensionless δ exhibits a significant increase, especially for the modal parameters of the first vertical bending and torsional modes (Fig. 15). It is difficult to stably identify the modal parameters during a high-wind period, which may be attributed to the structural unstable states. Referring to the ordinate value in Fig. 15, the identification uncertainty of the damping ratio is far greater than that of the structural frequency, particularly with respect to low damping modes (e.g., first-order vertical bending and torsional modes). Meanwhile, the damping ratio and the normalized standard deviation of the modal parameters for the first lateral bending mode from August 5 to 7, 2012 appear to display a certain correlation with the lateral acceleration RMS. The scatter plots of the modal frequency and the damping ratio versus the wind speed in Fig. 16 select the monitoring data according to the following two conditions to reduce the influence of the temperature and traffic load: (1) when Donghai Bridge was closed to traffic during four typhoon events (i.e., Krosa, Muifa, Haikui, and Bolaven); and (2) when the temperature at the top flange plate of the mid-span cross-section of the girder ranged from 20 °C to 25 °C. Clearly, the data points in Fig. 16 are quite scattered, and both the Pearson correlation coefficients R_{corr} are less than 0.1. Hence, the variation of the modal parameters exhibits no distinct patterns with the wind speed. Similarly, the correlation between the modal parameters and the acceleration RMS (even if the acceleration RMS in the low frequency band of 0–2.5 Hz was used) is statistically insignificant when the traffic load and temperature conditions are specified. Accordingly, the high wind can definitely cause changes in the structural modal parameters within a short time scale. However, the influencing mechanism underlying this phenomenon is very complex, and it is difficult to quantitatively explicate and describe the changes of the modal parameters merely through the wind speed or structural vibration intensity.

In addition, at the time of traffic resumption in Fig. 14, sudden changes occurs in the modal frequencies of Donghai Bridge in spite of the stable structural temperature and wind field. This result demonstrates that the traffic load has a non-negligible effect on the variation of the

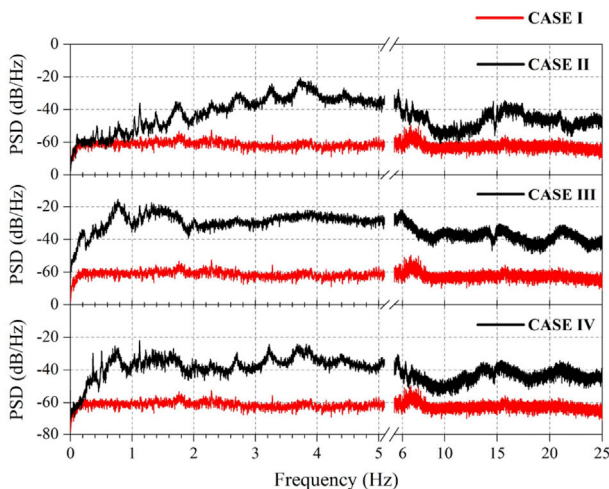


Fig. 12. PSD comparison of the lateral acceleration at the bottom of the north pylon.

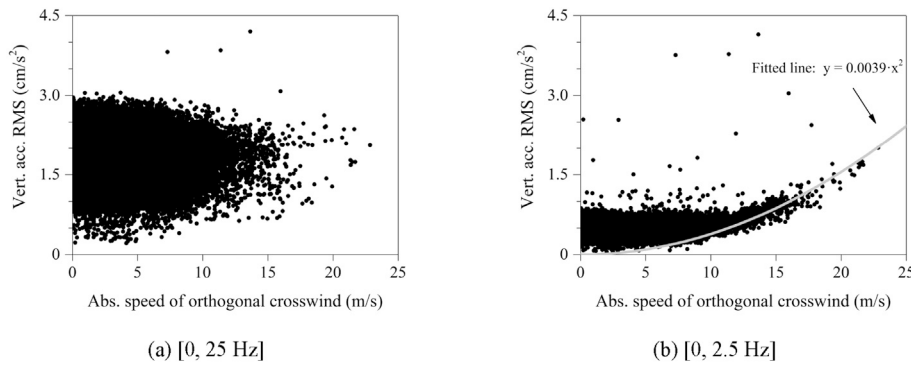


Fig. 13. Scatter plot of the RMS of acceleration in different frequency bands versus the speed of the orthogonal crosswind (2007–2012).

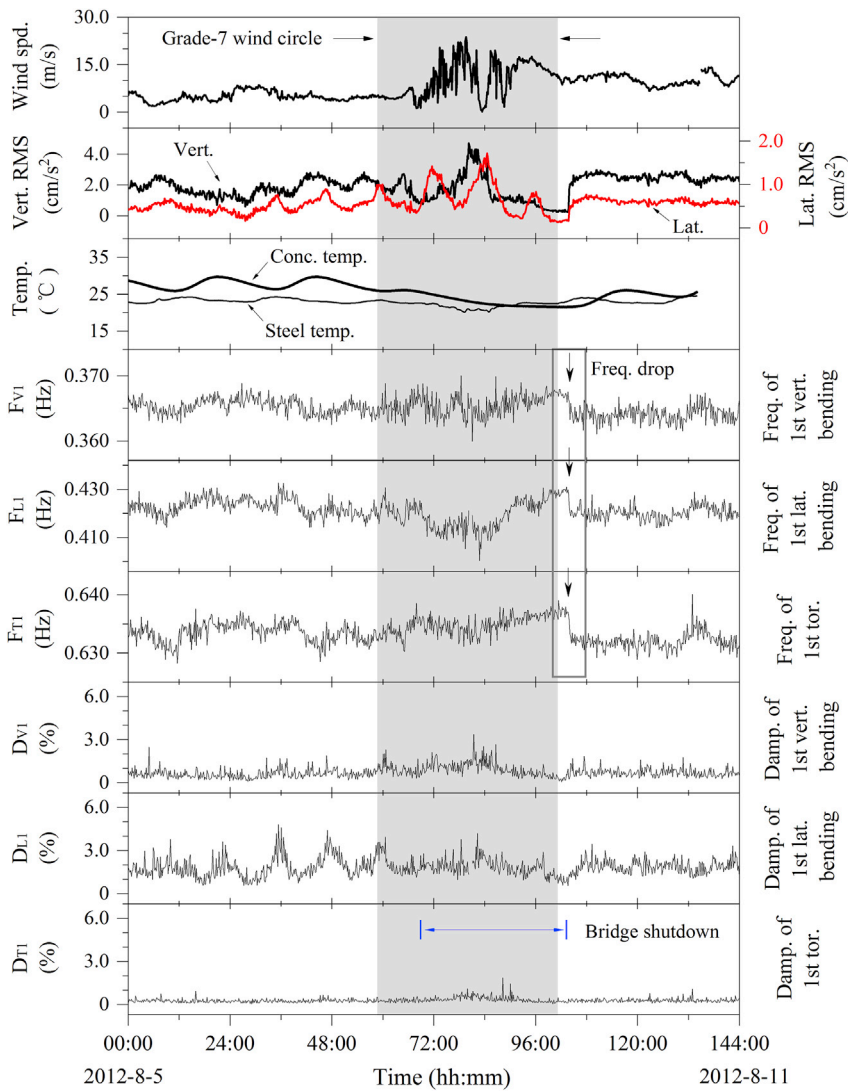


Fig. 14. Variations of the environmental factors and the modal parameters for Donghai Bridge during the Haikui typhoon event.

modal frequencies. Based on the periodic variability of the actions and modal frequencies of Donghai Bridge, we have further confirmed the influence of the temperature and traffic load on the frequency changes. The frequencies of the first-order vertical bending, lateral bending, and torsional modes of Donghai Bridge would decrease with the rising temperature or with the increasing traffic load. More details on this point can be found in (Zhou et al., 2017b).

6. Conclusions

The present study examines the wind-induced vibrational responses and the variations of the modal parameters during high-wind periods based on the SHM data of Donghai Bridge. A valuable comparison is made in terms of the respective contributions of high winds and traffics to the bridge vibration according to different wind and traffic scenarios. The following conclusions are drawn:

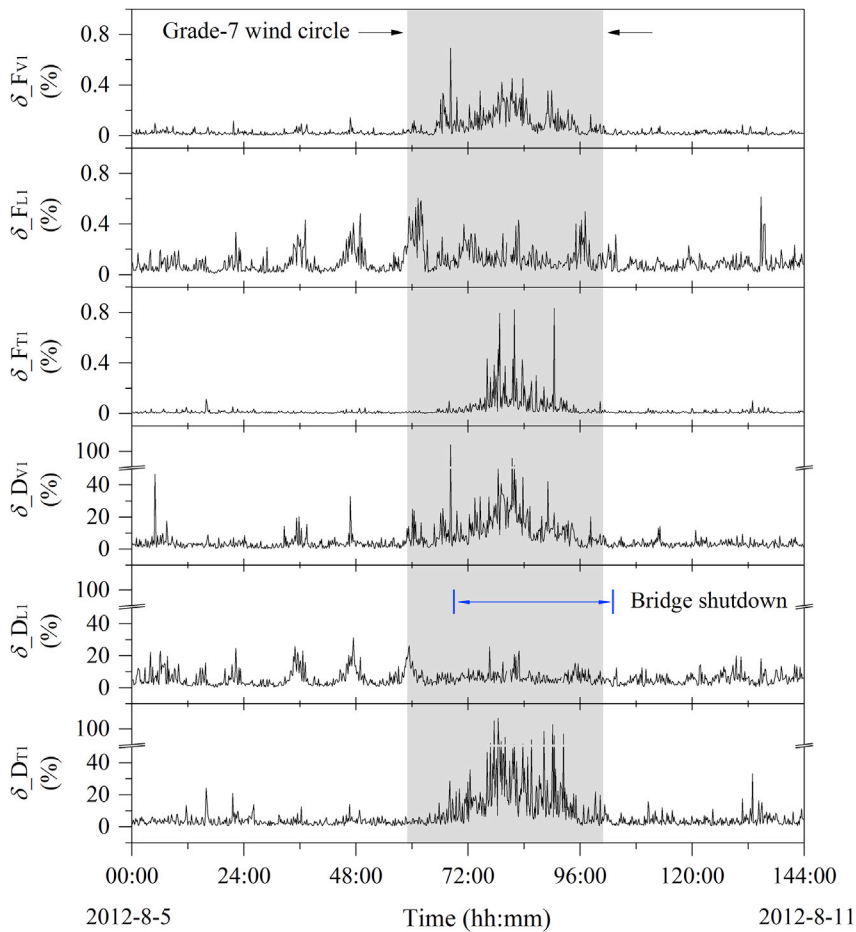


Fig. 15. Normalized standard deviations of the modal parameters for Donghai Bridge during the Haikui typhoon event.

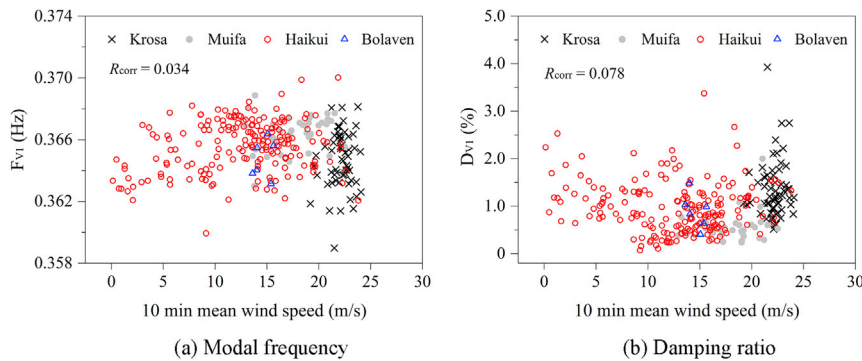


Fig. 16. Correlation between the modal parameter and the wind speed (first-order vertical bending mode).

- (1) The RMS of the wind-induced acceleration of Donghai Bridge increases with the mean wind speed in an approximately quadratic curve relation. Compared with the traffic load, high wind dominates the girder's lateral vibration of long-span bridges. However, for the vertical and torsional vibrations of the girder, heavy-traffic may exert more influence than the winds, such as the case in Donghai Bridge. According to the monitoring results, the frequency band of the wind excitation is lower than that of the traffic load, and the wind-induced vibration component can be highlighted through low-pass filtering.
- (2) During high winds, the modal parameters of a bridge are not easy to stably identify. Furthermore, their estimates suffer a relatively large uncertainty. The mechanism underlying the effects of high winds on the variations in the structural dynamic properties

appears to be quite complex, thereby presenting challenges in terms of the quantitative explication and depiction based on the wind speed or structural vibration intensity alone. After each high wind or typhoon event, the modal parameters of Donghai Bridge were able to recover to their states prior to the wind event, which indicated that its structural performance was not compromised by high winds.

This study could serve as a field evidence for the wind-resistant design and the performance evaluation of bridges in similar operational conditions.

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